

a 1983–2025, 20 milestones

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| 1983 IP ₃ releases Ca ²⁺ from an intracellular store | 2002 structure of the IP ₃ -binding core |
| 1989 primary structure — it is the receptor | 2007 <i>ITPR1</i> deletion causes SCA15 |
| 1989 purified receptor conducts Ca ²⁺ | 2014 <i>ITPR2</i> loss abolishes sweating |
| 1989 “similar to ryanodine receptor” | 2015 cryo-EM: the tetramer at 4.7 Å |
| 1990 P400 is the IP ₃ -binding protein | 2016 all four sites must be occupied |
| 1991 bell-shaped Ca ²⁺ dependence | 2018 paralogue structures with Ca ²⁺ and IP ₃ |
| 1991 subtypes differ by tissue and stage | 2020 <i>ITPR3</i> variants cause CMT1J |
| 1992 three receptors, not one | 2022 activation and gating across a ligand series |
| 1995 release is built from quantal puffs | 2024 <i>ITPR3</i> variant causes multisystem disease |
| 1996 <i>Itpr1</i> null mice are ataxic | 2025 recurrent CMT1J variant in nine families |

● what it is ● how it works ● what goes wrong

b the 137 references this review cites, and where the milestones fall

